

Lagrange Cardinal Polynomials and the Rvachev Up-Function

Closing an Exact Finite/Infinite Representation Loop

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Abstract

Let $u = \text{up}$ be Rvachev's compactly supported C^∞ function on $[-1, 1]$, normalized by $\int u = 1$, and let F be the Fabius function. This report gives two explicit representations of every Lagrange cardinal polynomial on $[-1, 1]$ by finitely many affine copies of u . The first is a transparent dyadic scale ladder with two atoms per degree; the second uses only translates at one common scale and admits the $d + 2$ -atom endpoint compression established in the repository's polynomial synthesis volume. For distinct nodes $\xi_0, \dots, \xi_d$, the scale-ladder coefficients are

$$a_{nj} = 2^{\binom{n+1}{2}} \sum_{r=0}^{d-n} \frac{\gamma_r}{r!} L_j^{(n+r)}(-2),$$

where L_j is the j 'th Lagrange cardinal polynomial and $\sum \gamma_r z^r / r! = 1/M_u(z)$ is the reciprocal moment-generating function of u . Thus Bernoulli cumulants and complete Bell polynomials enter the cardinal-to-atom transform explicitly.

Substitution into polynomial approximants closes two loops. First, every convergent Lagrange interpolation scheme for u becomes a limit of finite self-representations by shifted and scaled copies of u . Second, inserting the formula into the repository's absolutely and uniformly convergent Fourier–Legendre expansion gives, for every cutoff N , an exact finite identity

$$S_N(x) = \sum_{n=0}^{2N} A_{n,N} \left[u\left(-1 + \frac{x+3}{2^{n+1}}\right) + u\left(-1 + \frac{x+1}{2^{n+1}}\right) \right],$$

where $S_N = \sum_{m=0}^N \lambda_{2m} P_{2m}$ and $A_{n,N}$ is an explicit finite Bernoulli–Bell–Legendre transform. Since $S_N \rightarrow u$ uniformly, this is a closed Rvachev loop.

Several structural consequences are proved. The matrix $[G_n(\xi_i + 2)]$ is an Appell–Vandermonde matrix whose inverse columns are the Lagrange-to-up coefficients; its sample-to-atom determinant is

$$\det T_\Xi = \frac{2^{\binom{d+2}{3}} \prod_{n=0}^d n!}{\prod_{i < j} (\xi_j - \xi_i)}.$$

Consequently no choice of nodes removes the intrinsic quadratic-exponential coefficient amplification. A normal, coefficient-independent infinite series in these one-sided Appell blocks would make u analytic, so any fixed-coefficient closure must be strongly nonnormal. Moreover, the top atom coefficient in the degree- $2N$ Legendre loop has exact logarithmic limsup $2 \log 2$ after division by N^2 . Exact symbolic experiments through degree 32 verify all finite identities and show a largest amplitude about $10^{199.7}$ while the corresponding Legendre polynomial already has pointwise errors near 10^{-4} at 0 and 1.

The results described as new are new relative to the audited repository snapshot; no claim of global literature priority is made.

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1 Problem, terminology, and corpus audit

1.1 The requested loop

The problem has three logically distinct layers.

1. Start with a polynomial basis adapted to interpolation: for distinct nodes $\Xi_d = \{\xi_0, \dots, \xi_d\} \subset [-1, 1]$, write the cardinal polynomials

$$L_j(x) = \prod_{\substack{0 \leq m \leq d \\ m \neq j}} \frac{x - \xi_m}{\xi_j - \xi_m}, \quad L_j(\xi_i) = \delta_{ij}. \quad (1)$$

2. Represent each L_j exactly on $[-1, 1]$ by finitely many affine copies of $u = \text{up}$.
3. Insert those finite representations into an infinite polynomial representation of u and analyze the resulting self-referential limit.

A terminological distinction matters. The living primary exposition in the repository contains a Fourier–*Legendre* series for u , whereas the polynomials in (1) are *Lagrange* cardinal polynomials. These words are occasionally interchanged in informal requests, but they encode different constructions. This report closes both natural loops:

- a genuine Lagrange-interpolation loop, including a nested Lagrange–Newton series; and
- the substitution of the Lagrange/up formula into every partial sum of the existing Fourier–Legendre series.

1.2 Repository audit and duplication boundary

The recursive documentation tree was audited through its live directory map, archival inventory, draft manifest, and the current consolidated source volumes. The archive explicitly records historical versions and prunes byte-identical duplicates; the draft manifest records which earlier reports have been absorbed into canonical thematic volumes. The mathematical audit therefore emphasized the current sources while using the archive map to locate prior terminology and avoid silently rediscovering an archived result.

The closest antecedents are:

- the primary exposition *The Fabius Function and Rvachev’s Up-Function*, which gives the Fourier product, arithmetic dyadic values, moment formulas, asymptotics, nonanalyticity, and the absolutely and uniformly convergent Fourier–Legendre expansion [7];
- the consolidated *Representation Frontiers* volume, which collects transform, interpolation, sampling, and multiresolution representations [9];
- *Up-Function Polynomial Synthesis*, which proves exact finite polynomial reproduction by common-scale and scale-ladder dictionaries, the local dimension $d + 2$, and the minimal endpoint dictionary [8]; and
- the current manifest and archive README, which delimit the q-series, Lambert– W , Fourier-decay, integration, inverse-function, and Thue–Morse consolidations [10, 11].

The scale-ladder polynomial theorem and common-scale dictionary are therefore *corpus-known*. The following specializations and consequences were not found in the audited source text and are derived here:

1. an explicit closed formula for every Lagrange cardinal coefficient in terms of derivatives at -2 , reciprocal moments, Bernoulli numbers, and Bell polynomials;

2. the Appell–Vandermonde inverse and its exact sample-to-atom determinant;
3. node cancellation when a Lagrange representation of a polynomial is substituted into the up-atom ladder;
4. the direct closed Lagrange and Fourier–Legendre loops;
5. the normal-convergence obstruction and out-of-interval divergence mechanism;
6. the exact $2 \log 2$ quadratic growth law for the top Legendre-loop amplitude;
7. a prime-support theorem for rational equispaced cardinal amplitudes; and
8. a parity-adapted two-sided ladder and several frontier conjectures.

Remark 1.1 (Scope of the novelty statement). The repository is unusually large and was reorganized during August 2026. The audit used the canonical living volumes, recursive inventories, and full-text searches of the relevant consolidated sources. “New” below means new relative to that audited snapshot, not an assertion that no analogous formula exists anywhere in the global approximation-theory literature.

2 Fabius–Rvachev background

2.1 The up-function, the Fabius function, and the sinc product

Let $u = \text{up}$ be the unique normalized compactly supported solution of

$$u'(x) = 2(u(2x+1) - u(2x-1)), \quad \text{supp } u = [-1, 1], \quad \int_{\mathbb{R}} u(x) dx = 1. \quad (2)$$

It is even, nonnegative, and C^∞ . In the angular-frequency Fourier convention,

$$\widehat{u}(\omega) = \int_{\mathbb{R}} u(x) e^{-i\omega x} dx = \prod_{j=1}^{\infty} \frac{\sin(\omega/2^j)}{\omega/2^j}. \quad (3)$$

The Fabius function F is related on the support by

$$u(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1, \end{cases} \quad F(t) = u(t-1), \quad 0 \leq t \leq 1. \quad (4)$$

These normalizations agree with the standard arithmetic treatment of Arias de Reyna and with Rvachev’s survey [1, 2].

The product (3) gives the probabilistic model

$$Z = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad V_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1], \quad (5)$$

whose density is u . This model makes the moment calculus especially transparent.

2.2 Moments, Bernoulli cumulants, and Bell polynomials

Define

$$M(z) = \mathbb{E} e^{zZ} = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j} = \sum_{r=0}^{\infty} \mu_r \frac{z^r}{r!}. \quad (6)$$

The symmetry of Z gives $\mu_{2r+1} = 0$. The even cumulants are

$$\kappa_{2m} = \frac{B_{2m}}{2m(1 - 4^{-m})} = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad m \geq 1, \quad (7)$$

where B_{2m} are Bernoulli numbers. Consequently

$$\mu_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (8)$$

with $\mathcal{B}_n$ the complete exponential Bell polynomial.

The reciprocal moments γ_r are defined by

$$\frac{1}{M(z)} = \sum_{r=0}^{\infty} \gamma_r \frac{z^r}{r!}. \quad (9)$$

They satisfy

$$\gamma_n = \mathcal{B}_n(-\kappa_1, \dots, -\kappa_n), \quad \gamma_{2r+1} = 0. \quad (10)$$

The first values used later are shown in [Table 1](#).

Table 1: Moment and reciprocal-moment coefficients.

r	μ_r	γ_r
0	1	1
2	$\frac{1}{9}$	$-\frac{1}{9}$
4	$\frac{19}{675}$	$\frac{31}{675}$
6	$\frac{59535}{132809}$	$-\frac{2347}{1904389}$
8	$\frac{32531625}{46840699}$	$\frac{32531625}{3309760579}$
10	$\frac{24405225075}{4068990560161}$	$-\frac{24405225075}{1884491755011469}$
12	$\frac{4133856862760625}{4133856862760625}$	$\frac{4133856862760625}{4133856862760625}$

The moment–cumulant recurrence

$$a_n = \sum_{k=1}^n \binom{n-1}{k-1} c_k a_{n-k}, \quad a_0 = 1, \quad (11)$$

computes (μ_n) from $c_k = \kappa_k$ and (γ_n) from $c_k = -\kappa_k$. This is the exact recurrence used in the accompanying Python code.

2.3 The moment Appell family and its inverse

Define the Appell polynomials

$$G_n(y) = \mathbb{E}(y - Z)^n = M(D_y)y^n = \sum_{r=0}^n \binom{n}{r} \mu_r y^{n-r}. \quad (12)$$

Because Z is symmetric, $M(D_y)y^n = \mathbb{E}(y + Z)^n = \mathbb{E}(y - Z)^n$. The exponential generating function and Appell derivative law are

$$\sum_{n=0}^{\infty} G_n(y) \frac{z^n}{n!} = e^{yz} M(z), \quad G'_n(y) = n G_{n-1}(y). \quad (13)$$

The inverse differential operator is

$$\mathcal{D} := M(D_y)^{-1} = \sum_{r \geq 0} \gamma_r \frac{D_y^r}{r!}. \quad (14)$$

On a polynomial of degree d , the series terminates at $r = d$ and

$$M(D_y)\mathcal{D}p = p. \quad (15)$$

Thus the monomials and the G_n are connected by a finite triangular Bernoulli–Bell transform.

For orientation,

$$\begin{aligned} G_0(y) &= 1, \\ G_1(y) &= y, \\ G_2(y) &= y^2 + \frac{1}{9}, \\ G_3(y) &= y^3 + \frac{1}{3}y, \\ G_4(y) &= y^4 + \frac{2}{3}y^2 + \frac{19}{675}. \end{aligned} \quad (16)$$

3 The dyadic integration ladder and the two-atom blocks

This section gives a self-contained proof of the exact two-atom identity that drives the Lagrange synthesis.

3.1 One causal integration

Let

$$(Jf)(x) = \int_{-\infty}^x f(t) dt. \quad (17)$$

Lemma 3.1 (Causal dyadic integration). *For every $x \in \mathbb{R}$,*

$$Ju(x) = \sum_{k=0}^{\infty} u\left(\frac{x - (2k + 1)}{2}\right). \quad (18)$$

The sum is locally finite.

Proof. Set the right-hand side equal to $H(x)$. Compact support makes the sum locally finite. Differentiating and using (2),

$$\begin{aligned} H'(x) &= \frac{1}{2} \sum_{k \geq 0} u'\left(\frac{x - (2k + 1)}{2}\right) \\ &= \sum_{k \geq 0} (u(x - 2k) - u(x - 2k - 2)) = u(x), \end{aligned}$$

by telescoping. Both $H(x)$ and $Ju(x)$ tend to zero as $x \rightarrow -\infty$, so they are equal. $\square$

3.2 Repeated integration and binary partitions

For $m \geq 1$, define $p_m(N)$ by

$$\sum_{N=0}^{\infty} p_m(N) q^N = \prod_{\ell=0}^{m-1} \frac{1}{1 - q^{2^\ell}}. \quad (19)$$

Thus $p_m(N)$ counts partitions of N into powers $1, 2, \dots, 2^{m-1}$, with repetitions allowed. In particular, $p_m(0) = p_m(1) = 1$.

Lemma 3.2 (Repeated dyadic integration). *For $m \geq 1$,*

$$J^m u(y) = 2^{\binom{m}{2}} \sum_{N=0}^{\infty} p_m(N) u\left(\frac{y - (2^m - 1 + 2N)}{2^m}\right). \quad (20)$$

Proof. The case $m = 1$ is (18), since $p_1(N) = 1$. Suppose the formula holds for m . Integrating one scaled summand and applying (18) gives

$$\begin{aligned} & \int_{-\infty}^y u\left(\frac{t - (2^m - 1 + 2N)}{2^m}\right) dt \\ &= 2^m \sum_{k \geq 0} u\left(\frac{y - (2^{m+1} - 1 + 2(N + 2^m k))}{2^{m+1}}\right). \end{aligned}$$

Hence the new coefficient of index R is $\sum_{k \geq 0} p_m(R - 2^m k)$, whose generating function is the old one divided by $1 - q^{2^m}$. The prefactor is multiplied by 2^m , changing $2^{\binom{m}{2}}$ to $2^{\binom{m+1}{2}}$. $\square$

The reciprocal lacunary product in (19) is the integration-side companion of the finite Thue–Morse product

$$\Theta_m(q) = \prod_{\ell=0}^{m-1} (1 - q^{2^\ell}) = \sum_{r=0}^{2^m-1} (-1)^{s_2(r)} q^r. \quad (21)$$

This is the first direct bridge between the present interpolation problem and the Thue–Morse/q-product layer of the repository.

3.3 Tail moments and the two-atom theorem

For $y \geq 1$, the entire support of u lies below y , so

$$J^{n+1} u(y) = \frac{1}{n!} \int_{-1}^1 (y - t)^n u(t) dt = \frac{G_n(y)}{n!}. \quad (22)$$

Combining this with (20) is decisive.

Theorem 3.3 (Two-atom Appell block on $-1 \leq x \leq 1$). *For $n \geq 0$ define*

$$U_n(x) = u\left(-1 + \frac{x+3}{2^{n+1}}\right) + u\left(-1 + \frac{x+1}{2^{n+1}}\right), \quad (23)$$

$$D_n = n! 2^{\binom{n+1}{2}}. \quad (24)$$

Then, for every $x \in [-1, 1]$,

$$\boxed{D_n U_n(x) = G_n(x+2)}. \quad (25)$$

In particular, $D_n U_n$ is a monic polynomial of degree n on the interval.

Proof. Put $m = n + 1$ and $y = x + 2 \in [1, 3]$ in (20). If $N \geq 2$, then

$$\frac{y - (2^{n+1} - 1 + 2N)}{2^{n+1}} \leq -1,$$

with equality only at $y = 3$, $N = 2$, where $u(-1) = 0$. Thus only $N = 0$ and $N = 1$ survive, and both have coefficient 1. Equation (22) then yields

$$\frac{G_n(y)}{n!} = 2^{\binom{n+1}{2}} \left[u\left(\frac{y - (2^{n+1} - 1)}{2^{n+1}}\right) + u\left(\frac{y - (2^{n+1} + 1)}{2^{n+1}}\right) \right].$$

Substituting $y = x + 2$ gives (25). $\square$

Corollary 3.4 (Fabius two-ray transfer identity). *For $n \geq 1$ and $y \in [1, 3]$,*

$$F\left(\frac{y+1}{2^{n+1}}\right) + F\left(\frac{y-1}{2^{n+1}}\right) = \frac{G_n(y)}{n!2^{\binom{n+1}{2}}}. \quad (26)$$

At $y = 1$ this gives the reciprocal-moment identity

$$\boxed{F(2^{-n}) = \frac{G_n(1)}{n!2^{\binom{n+1}{2}}}.} \quad (27)$$

Consequently

$$\frac{F((y+1)2^{-n-1}) + F((y-1)2^{-n-1})}{F(2^{-n})} = \frac{G_n(y)}{G_n(1)}. \quad (28)$$

Proof. For $n \geq 1$, both arguments of u in (23) lie in $[-1, 0]$. Apply (4). At $y = 1$, the second Fabius argument is zero and $F(0) = 0$. $\square$

For $n = 0$, (25) is the partition identity

$$u\left(\frac{x+1}{2}\right) + u\left(\frac{x-1}{2}\right) = 1, \quad -1 \leq x \leq 1. \quad (29)$$

4 Exact Lagrange cardinal synthesis

4.1 The main finite formula

Let $\Xi_d = \{\xi_0, \dots, \xi_d\} \subset [-1, 1]$ consist of distinct nodes, and let L_j be defined by (1).

Theorem 4.1 (Lagrange cardinal as a finite up ladder). *For $0 \leq j \leq d$, define*

$$b_{nj} = \frac{1}{n!} \sum_{r=0}^{d-n} \frac{\gamma_r}{r!} L_j^{(n+r)}(-2), \quad 0 \leq n \leq d, \quad (30)$$

$$a_{nj} = D_n b_{nj} = 2^{\binom{n+1}{2}} \sum_{r=0}^{d-n} \frac{\gamma_r}{r!} L_j^{(n+r)}(-2). \quad (31)$$

Then, exactly on $[-1, 1]$,

$$L_j(x) = \sum_{n=0}^d b_{nj} G_n(x+2) \quad (32)$$

$$= \boxed{\sum_{n=0}^d a_{nj} \left[u\left(-1 + \frac{x+3}{2^{n+1}}\right) + u\left(-1 + \frac{x+1}{2^{n+1}}\right) \right]}. \quad (33)$$

Thus a degree- d cardinal polynomial uses at most $2(d+1)$ affine up-atoms, organized as $d+1$ equal-coefficient pairs.

Proof. Set $q_j(y) = L_j(y-2)$. Apply the finite inverse operator (14):

$$B_j(y) := \mathcal{D}q_j(y) = M(D_y)^{-1}q_j(y) = \sum_{n=0}^d b_{nj} y^n. \quad (34)$$

The coefficient of y^n in the r th derivative of q_j is $q_j^{(n+r)}(0)/n! = L_j^{(n+r)}(-2)/n!$, proving (30). Now

$$q_j(y) = M(D_y)B_j(y) = \sum_{n=0}^d b_{nj} G_n(y).$$

Set $y = x+2$ to obtain (32), then invoke Theorem 3.3 to obtain (33). $\square$

Because odd γ_r vanish, (31) can be shortened to

$$a_{nj} = 2^{\binom{n+1}{2}} \sum_{s=0}^{\lfloor (d-n)/2 \rfloor} \frac{\gamma_{2s}}{(2s)!} L_j^{(n+2s)}(-2). \quad (35)$$

Equations (7), (10), and (35) are the promised Bernoulli–Bell formula for the Lagrange-to-Rvachev transform.

4.2 A nodes-only elementary-symmetric formula

The derivatives at -2 can be computed without first expanding L_j . Define

$$\rho_{jm} = \frac{1}{-2 - \xi_m}, \quad m \neq j, \quad (36)$$

and let $e_q^{(j)}$ be the elementary symmetric polynomial of degree q in the d numbers $\{\rho_{jm} : m \neq j\}$. Since

$$L_j(-2 + z) = L_j(-2) \prod_{m \neq j} (1 + \rho_{jm} z), \quad (37)$$

we have

$$L_j^{(q)}(-2) = q! L_j(-2) e_q^{(j)}. \quad (38)$$

Corollary 4.2 (Explicit nodes-only amplitude). *The amplitude in (33) is*

$$a_{nj} = 2^{\binom{n+1}{2}} L_j(-2) \sum_{s=0}^{\lfloor (d-n)/2 \rfloor} \gamma_{2s} \frac{(n+2s)!}{(2s)!} e_{n+2s}^{(j)}. \quad (39)$$

Here

$$L_j(-2) = \prod_{m \neq j} \frac{-2 - \xi_m}{\xi_j - \xi_m}. \quad (40)$$

This form separates three structures: the interpolation geometry in $L_j(-2)$ and $e_q^{(j)}$, the Rvachev law in γ_{2s} , and the dyadic integration normalization in $2^{\binom{n+1}{2}}$.

4.3 The Appell–Vandermonde inverse

Define the $(d+1) \times (d+1)$ matrix

$$A_{\Xi} = (A_{in})_{0 \leq i, n \leq d}, \quad A_{in} = G_n(\xi_i + 2). \quad (41)$$

Theorem 4.3 (Appell–Vandermonde inversion). *Let $B = (b_{nj})_{0 \leq n, j \leq d}$ with b_{nj} from (30). Then*

$$B = A_{\Xi}^{-1}. \quad (42)$$

Moreover,

$$\det A_{\Xi} = \prod_{0 \leq i < j \leq d} (\xi_j - \xi_i). \quad (43)$$

Proof. Evaluating (32) at $x = \xi_i$ gives

$$\delta_{ij} = \sum_{n=0}^d A_{in} b_{nj},$$

so $A_{\Xi} B = I$. Each $G_n(y)$ is monic of degree n . Changing from the monomial columns $(1, y, \dots, y^d)$ to $(G_0(y), \dots, G_d(y))$ is unit triangular, hence does not change the Vandermonde determinant. Replacing $y_i = \xi_i + 2$ by ξ_i also leaves all pairwise differences unchanged. $\square$

For sample data $\mathbf{f} = (f(\xi_0), \dots, f(\xi_d))^\top$, the Appell coordinates are $\mathbf{c} = \mathbf{A}_\Xi^{-1} \mathbf{f}$, while the raw two-atom amplitudes are

$$\mathbf{A} = \mathbf{T}_\Xi \mathbf{f}, \quad \mathbf{T}_\Xi = \text{diag}(D_0, \dots, D_d) \mathbf{A}_\Xi^{-1}. \quad (44)$$

Corollary 4.4 (Exact determinant of the sample-to-atom transform). *For ordered distinct nodes,*

$$\det \mathbf{T}_\Xi = \frac{2^{\binom{d+2}{3}} \prod_{n=0}^d n!}{\prod_{0 \leq i < j \leq d} (\xi_j - \xi_i)}. \quad (45)$$

Proof. Multiply (43) by $\prod_n D_n$ and use

$$\sum_{n=0}^d \binom{n+1}{2} = \binom{d+2}{3}.$$

□

4.4 A worked quadratic cardinal system

Take $\Xi_2 = \{-1, 0, 1\}$. The exact coefficient table is Table 2.

Table 2: Quadratic Lagrange cardinals in the Appell basis and in raw up-pair amplitudes.

j	$L_j(x)$	b_{0j}	b_{1j}	b_{2j}	α_{0j}	α_{1j}	α_{2j}
0	$\frac{x(x-1)}{2}$	$\frac{53}{18}$	$-\frac{5}{2}$	$\frac{1}{2}$	$\frac{53}{18}$	-5	8
1	$-(x-1)(x+1)$	$-\frac{26}{9}$	4	-1	$-\frac{26}{9}$	8	-16
2	$\frac{x(x+1)}{2}$	$\frac{17}{18}$	$-\frac{3}{2}$	$\frac{1}{2}$	$\frac{17}{18}$	-3	8

Writing

$$\begin{aligned} U_0(x) &= u\left(\frac{x+1}{2}\right) + u\left(\frac{x-1}{2}\right), \\ U_1(x) &= u\left(\frac{x-1}{4}\right) + u\left(\frac{x-3}{4}\right), \\ U_2(x) &= u\left(\frac{x-5}{8}\right) + u\left(\frac{x-7}{8}\right), \end{aligned}$$

we obtain, on $[-1, 1]$,

$$\frac{x(x-1)}{2} = \frac{53}{18}U_0(x) - 5U_1(x) + 8U_2(x), \quad (46)$$

$$1 - x^2 = -\frac{26}{9}U_0(x) + 8U_1(x) - 16U_2(x), \quad (47)$$

$$\frac{x(x+1)}{2} = \frac{17}{18}U_0(x) - 3U_1(x) + 8U_2(x). \quad (48)$$

All three cardinals share the same six-atom dictionary.

4.5 Algorithm and complexity

Algorithm 4.5 (Exact Lagrange-to-up transform). Given distinct nodes $\xi_0, \dots, \xi_d$:

1. Compute κ_{2m} from (7) for $1 \leq m \leq \lfloor d/2 \rfloor$.
2. Compute $\gamma_0, \dots, \gamma_d$ by the recurrence (11) with cumulants $-\kappa_r$.
3. For each j , form $L_j(-2)$ and the elementary symmetric arrays $e_q^{(j)}$ from (36)–(37).
4. Evaluate (39) for $0 \leq n \leq d$.
5. Emit the two affine atoms in (23), each with amplitude a_{nj} .

A straightforward implementation costs $\mathcal{O}(d^3)$ arithmetic operations for all $d+1$ cardinals. Product trees or prefix/suffix elementary-symmetric updates reduce the structured part to $\mathcal{O}(d^2)$. For rational nodes all operations can be exact. Floating-point computation should use high precision because the anchor -2 lies outside the interpolation interval and the factors D_n grow superexponentially in n .

5 A common-scale representation using shifts only

The scale ladder in Theorem 4.1 changes the dilation with n . If “shifted copies” is interpreted literally as a fixed dilation with only translations varying, the repository’s common-scale theorem gives a second answer.

5.1 Global common-scale synthesis

Let $P \in \Pi_d$ be a polynomial on $[0, 1]$, put $H = 2^d$, and retain $M(z)$ for the moment-generating function in (6). Define

$$Q := M(HD_t)^{-1}P = \sum_{r=0}^d \gamma_r H^r \frac{P^{(r)}}{r!}. \quad (49)$$

The common-scale reproduction theorem from [8] states

$$P(t) = \frac{1}{H} \sum_{k \in \mathbb{Z}} Q(k) u\left(\frac{t-k}{H}\right). \quad (50)$$

On $[0, 1]$ only $-H+1 \leq k \leq H$ contribute.

For the cardinal L_j on $[-1, 1]$, set

$$P_j(t) = L_j(2t-1). \quad (51)$$

Then (50) becomes

$$L_j(x) = \frac{1}{H} \sum_{k=-H+1}^H Q_j(k) u\left(\frac{x+1-2k}{2H}\right), \quad -1 \leq x \leq 1, \quad (52)$$

where

$$Q_j(t) = \sum_{r=0}^d \gamma_r H^r \frac{P_j^{(r)}(t)}{r!}. \quad (53)$$

All $2H = 2^{d+1}$ atoms have the same dilation $2H$ in the x variable.

The scale $H = 2^d$ is minimal among positive integer cardinal scales. In the angular-frequency convention of (3), the minimum multiplicity of a lattice zero $\hat{u}(2\pi H\ell)$ over $\ell \in \mathbb{Z} \setminus \{0\}$ is $v_2(H) + 1$. Reproduction of all polynomials through degree d requires multiplicity at least $d+1$, hence $v_2(H) \geq d$ and the smallest integer choice is $H = 2^d$. This is a sharp Strang–Fix phenomenon, consistent with the classical shift-invariant framework [3, 4].

5.2 The minimal $d+2$ endpoint dictionary

The common-scale representation has a large alias nullspace. Define

$$A_d(z) = \prod_{\ell=1}^d (z^{2^{\ell-1}} + 1)^{d-\ell+1} = \prod_{m=1}^d [2^m]_z, \quad (54)$$

where $[N]_z = (1 - z^N)/(1 - z)$. Its degree is $2^{d+1} - d - 2$, and the identity

$$A_d(z)(z - 1)^{d+1} = (-1)^{d+1}(1 - z)\Theta_d(z^2) \quad (55)$$

links the alias annihilator directly to the finite Thue–Morse product.

The endpoint-basis theorem of [8] compresses (52) to the $d + 2$ rightmost centers. In the present coordinates there are unique coefficients β_{jk} , rational whenever the nodes are rational, for $H - d - 1 \leq k \leq H$, such that

$$L_j(x) = \frac{1}{H} \sum_{k=H-d-1}^H \beta_{jk} u\left(\frac{x+1-2k}{2H}\right), \quad -1 \leq x \leq 1. \quad (56)$$

They are obtained by extending the early values $Q_j(k)$ through the monic recurrence $A_d(E)\nu = 0$ and subtracting the resulting null sequence on the last $d + 2$ centers.

Remark 5.1 (Two exact answers to the finite-synthesis problem). The ladder (33) uses only $2(d + 1)$ atoms and has a closed coefficient formula, but it spans exponentially separated scales. The endpoint representation (56) uses the optimal fixed dictionary size $d + 2$ and one common scale, but its scale is 2^d and its coefficient extraction is governed by a high-order alias recurrence. These are complementary, not competing, normal forms.

6 The first closed loop: Lagrange interpolation of the up-function

6.1 Finite interpolation and atom synthesis

For a function f on $[-1, 1]$, let

$$(I_{\Xi_d} f)(x) = \sum_{j=0}^d f(\xi_j) L_j(x). \quad (57)$$

Substitute (33) and define

$$c_{n,d}(f) = \sum_{j=0}^d f(\xi_j) b_{nj}, \quad (58)$$

$$A_{n,d}(f) = D_n c_{n,d}(f) = \sum_{j=0}^d f(\xi_j) a_{nj}. \quad (59)$$

Then

$$I_{\Xi_d} f(x) = \sum_{n=0}^d c_{n,d}(f) G_n(x+2) = \sum_{n=0}^d A_{n,d}(f) U_n(x). \quad (60)$$

For $f = u$, this is already a finite self-representation: values of u at the interpolation nodes determine a finite sum of shifted and scaled copies of u that agrees with the interpolating polynomial on the entire interval.

In matrix form,

$$\mathbf{c} = \mathbf{A}_{\Xi}^{-1} \mathbf{f}, \quad \mathbf{A} = \mathbf{T}_{\Xi} \mathbf{f}. \quad (61)$$

Thus the Lagrange basis is an intermediate factorization of the unique Appell coordinates.

6.2 Uniform convergence on Chebyshev–Lobatto grids

Let

$$\xi_j^{(d)} = -\cos \frac{j\pi}{d}, \quad 0 \leq j \leq d, \quad (62)$$

with the usual interpretation for $d = 0$. Their Lebesgue constants satisfy $\Lambda_d = \mathcal{O}(\log d)$. If $E_d(f)$ denotes best uniform polynomial approximation error, then

$$\|f - I_d f\|_\infty \leq (1 + \Lambda_d) E_d(f). \quad (63)$$

Because $u \in C^\infty[-1, 1]$, Jackson estimates give, for every r ,

$$E_d(u) = \mathcal{O}_r(d^{-r}). \quad (64)$$

Theorem 6.1 (Closed Lagrange interpolation loop). *For Chebyshev–Lobatto nodes, let $A_{n,d} = A_{n,d}(u)$ be given by (59). Then*

$$u(x) = \lim_{d \rightarrow \infty} \sum_{n=0}^d A_{n,d} \left[u\left(-1 + \frac{x+3}{2^{n+1}}\right) + u\left(-1 + \frac{x+1}{2^{n+1}}\right) \right] \quad (65)$$

uniformly for $x \in [-1, 1]$. The convergence is faster than every negative power of d . For each fixed derivative order s , the same conclusion holds in $C^s[-1, 1]$ after taking sufficiently high smoothness in the Jackson–Markov estimate.

Proof. The d th finite sum in (65) is exactly $I_d u$ by (60). Equations (63) and (64) prove the uniform claim. For derivatives, approximate u by a polynomial with sufficiently small C^s error and apply Markov’s inequality to the interpolated remainder; since u has derivatives of every order, the algebraic power can be chosen arbitrarily large. $\square$

6.3 Nested Lagrange–Newton series

A nested node sequence $\xi_0, \xi_1, \dots$ gives the normalized Newton factors

$$N_0(x) = 1, \quad N_d(x) = \prod_{m=0}^{d-1} \frac{x - \xi_m}{\xi_d - \xi_m}, \quad d \geq 1. \quad (66)$$

Let

$$\Delta_d f = f(\xi_d) - I_{d-1} f(\xi_d), \quad I_{-1} f = 0. \quad (67)$$

Then

$$I_D f(x) = \sum_{d=0}^D \Delta_d f N_d(x). \quad (68)$$

Each N_d is a degree- d Lagrange cardinal polynomial for the augmented node set and therefore has an expansion

$$N_d(x) = \sum_{n=0}^d \alpha_{n,d} U_n(x) \quad (69)$$

from Theorem 4.1. Consequently

$$I_D u(x) = \sum_{n=0}^D \left(\sum_{d=n}^D \Delta_d u \alpha_{n,d} \right) U_n(x). \quad (70)$$

For a convergent nested interpolation sequence, such as a suitable Leja ordering, (70) is a genuine triangular Lagrange series closed through finite up-atom blocks. The inner coefficient depends on the cutoff D ; Section 8.5 explains why one should not expect a stable, normally convergent cutoff-independent limit.

6.4 A compactly supported smooth extension operator

The right-hand side of (33) makes sense globally and is $C_c^\infty(\mathbb{R})$. Let $\tilde{L}_j$ denote that global atom sum. Then

$$\tilde{L}_j|_{[-1,1]} = L_j. \quad (71)$$

For nodal data $\mathbf{f} = (f_0, \dots, f_d)^\top$, define

$$(\tilde{\mathcal{E}}_d \mathbf{f})(x) = \sum_{j=0}^d f_j \tilde{L}_j(x). \quad (72)$$

This is a compactly supported C^∞ cardinal extension. Equivalently, define $\mathcal{E}_d : \Pi_d \rightarrow C_c^\infty(\mathbb{R})$ by $\mathcal{E}_d p = \tilde{\mathcal{E}}_d(p(\xi_0), \dots, p(\xi_d))^\top$. If R denotes restriction to $[-1, 1]$ and I_d polynomial interpolation, then

$$\mathcal{P}_d = \mathcal{E}_d I_d R \quad (73)$$

is idempotent on its natural global function space because $R\mathcal{E}_d = \text{Id}_{\Pi_d}$. The closed loop is therefore a finite-rank projection conjugated through a smooth atomic extension.

7 The second closed loop: the Fourier–Legendre expansion

7.1 The corpus-known Legendre series

Let P_m denote the Legendre polynomial normalized by $P_m(1) = 1$. Since u is even, its Fourier–Legendre expansion has the form

$$u(x) = \sum_{m=0}^{\infty} \lambda_{2m} P_{2m}(x), \quad \lambda_{2m} = \frac{4m+1}{2} \int_{-1}^1 u(x) P_{2m}(x) dx. \quad (74)$$

The primary repository exposition proves absolute and uniform convergence on $[-1, 1]$ and gives the exact dyadic formula

$$\lambda_{2m} = 4^{-m} (4m+1) \sum_{k=0}^m (-1)^{m+k} \binom{2m}{m+k} \binom{2m+2k}{2m} (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (75)$$

See [7]. Write

$$S_N(x) = \sum_{m=0}^N \lambda_{2m} P_{2m}(x). \quad (76)$$

7.2 Literal substitution through Lagrange cardinals

Choose any $2N+1$ distinct nodes $\Xi_{2N} \subset [-1, 1]$. Since $S_N \in \Pi_{2N}$,

$$S_N(x) = \sum_{j=0}^{2N} S_N(\xi_j) L_j(x). \quad (77)$$

Substituting (33) gives

$$S_N(x) = \sum_{n=0}^{2N} \left(\sum_{j=0}^{2N} S_N(\xi_j) a_{nj} \right) U_n(x). \quad (78)$$

At first sight the coefficients depend on the arbitrary interpolation grid. They do not.

Theorem 7.1 (Node cancellation). *For every polynomial $p \in \Pi_d$ and every distinct node set Ξ_d ,*

$$\sum_{j=0}^d p(\xi_j) b_{nj} = [y^n] M(D_y)^{-1} p(y-2), \quad 0 \leq n \leq d. \quad (79)$$

Hence the up-block coefficients obtained by passing through any Lagrange grid are independent of that grid.

Proof. The sample vector of p is $\mathbf{p} = \mathbf{A}_\Xi \mathbf{c}$, where $\mathbf{c}$ is the unique Appell coordinate vector of p . Multiplication by $\mathbf{B} = \mathbf{A}_\Xi^{-1}$ recovers $\mathbf{c}$. The differential formula for $\mathbf{c}$ is exactly (79). $\square$

Thus Lagrange interpolation is not an extra approximation at a fixed Legendre cutoff; it is an exact factorization of the same triangular Appell transform.

7.3 The explicit Legendre-to-up transform

Define

$$c_{n,N} = [y^n] M(D_y)^{-1} S_N(y-2) \quad (80)$$

$$= \frac{1}{n!} \sum_{r=0}^{2N-n} \frac{\gamma_r}{r!} S_N^{(n+r)}(-2), \quad (81)$$

$$A_{n,N} = D_n c_{n,N}. \quad (82)$$

Then Theorem 3.3 gives an exact identity for every N .

Theorem 7.2 (Finite Fourier–Legendre/up closure). *For $-1 \leq x \leq 1$,*

$$S_N(x) = \sum_{n=0}^{2N} A_{n,N} \left[u \left(-1 + \frac{x+3}{2^{n+1}} \right) + u \left(-1 + \frac{x+1}{2^{n+1}} \right) \right]. \quad (83)$$

The coefficient is the finite transform

$$A_{n,N} = 2^{\binom{n+1}{2}} \sum_{m=\lceil n/2 \rceil}^N \lambda_{2m} \sum_{r=0}^{2m-n} \frac{\gamma_r}{r!} P_{2m}^{(n+r)}(-2). \quad (84)$$

Only even r contribute. Inserting (75) expresses every $A_{n,N}$ as a finite sum of Bernoulli–Bell coefficients and dyadic Fabius values.

Proof. Apply (15) to the polynomial $S_N(y-2)$, then apply $M(D_y)$ and the basis identity (12). This gives $S_N(x) = \sum c_{n,N} G_n(x+2)$. The two-atom theorem gives (83). Expanding S_N in (81) gives (84). $\square$

The derivatives in (84) may be written with Gegenbauer polynomials:

$$P_m^{(q)}(z) = 2^q \left(\frac{1}{2} \right)_q C_{m-q}^{q+1/2}(z), \quad q \leq m. \quad (85)$$

Thus the loop can be made completely explicit in classical special functions.

Corollary 7.3 (Infinite closed Fourier–Legendre loop). *The limit*

$$u(x) = \lim_{N \rightarrow \infty} \sum_{n=0}^{2N} A_{n,N} U_n(x) \quad (86)$$

converges absolutely through the Legendre side and uniformly in x on $[-1, 1]$.

Proof. The finite sum equals S_N identically by Theorem 7.2, and $S_N \rightarrow u$ uniformly by (74). $\square$

7.4 The first three finite loops

The first Legendre coefficients are

$$\lambda_0 = \frac{1}{2}, \quad \lambda_2 = -\frac{5}{6}, \quad \lambda_4 = \frac{11}{30}, \quad \lambda_6 = \frac{143}{5670}. \quad (87)$$

The first partial sums and their atom closures are

$$S_0(x) = \frac{1}{2}U_0(x), \quad (88)$$

$$S_1(x) = -\frac{71}{18}U_0(x) + 10U_1(x) - 20U_2(x), \quad (89)$$

$$S_2(x) = \frac{49849}{4050}U_0(x) - \frac{658}{9}U_1(x) + \frac{5012}{9}U_2(x) - 4928U_3(x) + 39424U_4(x). \quad (90)$$

Already at degree four, exact cancellation involves amplitudes of order 10^4 .

8 Structural analysis of the closed loop

8.1 Finite-rank conjugacy

Let $\mathcal{S}_d : \Pi_d \rightarrow C_c^\infty(\mathbb{R})$ denote the global ladder extension

$$(\mathcal{S}_d p)(x) = \sum_{n=0}^d D_n[y^n] M(D_y)^{-1} p(y-2) U_n(x). \quad (91)$$

Then

$$(\mathcal{S}_d p)|_{[-1,1]} = p. \quad (92)$$

The Lagrange and Legendre closures are respectively

$$\mathcal{S}_d I_{\Xi_d} u, \quad \mathcal{S}_{2N} S_N, \quad (93)$$

restricted back to $[-1, 1]$. The “loop” is therefore an exact right inverse of restriction composed with a sequence of finite-rank polynomial projectors.

8.2 Intrinsic sample-to-atom amplification

Equation (45) immediately forces large coefficient maps.

Theorem 8.1 (Node-independent lower bound). *For every distinct node set $\Xi_d \subset [-1, 1]$,*

$$\|\mathbf{T}_\Xi\|_2 \geq 2^{d(d-1)/6} \left(\prod_{n=0}^d n! \right)^{1/(d+1)}. \quad (94)$$

More precisely, Stirling–Barnes asymptotics give

$$\log \|\mathbf{T}_\Xi\|_2 \geq \frac{\log 2}{6} d^2 + \frac{1}{2} d \log d - \left(\frac{\log 2}{6} + \frac{3}{4} \right) d + \mathcal{O}(\log d). \quad (95)$$

No choice of interpolation nodes can make the raw ladder transform uniformly well conditioned as $d \rightarrow \infty$.

Proof. For a square matrix of size $d + 1$, $\|T\|_2^{d+1} \geq |\det T|$. Since $|\xi_j - \xi_i| \leq 2$,

$$\left| \prod_{i < j} (\xi_j - \xi_i) \right| \leq 2^{\binom{d+1}{2}}.$$

Insert this into (45) and use

$$\binom{d+2}{3} - \binom{d+1}{2} = \binom{d+1}{3}.$$

Taking the $(d + 1)$ st root gives (94). $\square$

Chebyshev or Fekete nodes improve the Vandermonde factor, but they cannot remove the dominant $2^{\binom{d+2}{3}}$ volume expansion. This distinction is visible in the numerical experiments of Section 11.

8.3 Why the anchor minus two is unstable

The transform (81) samples S_N and its derivatives at -2 , outside the approximation interval. The Legendre polynomial has the classical exterior growth

$$|P_m(-2)| \asymp \frac{(2 + \sqrt{3})^m}{\sqrt{m}}. \quad (96)$$

Let $\rho_* = 2 + \sqrt{3}$.

Theorem 8.2 (Termwise divergence at the Appell anchor). *The terms of the Legendre series for u do not tend to zero at $x = -2$. More strongly, $|\lambda_{2m} P_{2m}(-2)|$ is unbounded along a subsequence.*

Proof. If $|\lambda_{2m}| = \mathcal{O}(\rho^{-2m})$ for some $\rho > 1$, then the Legendre series converges normally in every smaller Bernstein ellipse and defines an analytic continuation of u . Since u is nowhere analytic on its support [7], no such ρ exists. Fix $1 < \rho < \rho_*$. Infinitely many m satisfy $|\lambda_{2m}| > \rho^{-2m}$. Along that subsequence,

$$|\lambda_{2m} P_{2m}(-2)| \gtrsim \frac{(\rho_*/\rho)^{2m}}{\sqrt{m}} \rightarrow \infty$$

by (96). $\square$

This theorem explains why an excellent approximation on $[-1, 1]$ can become a huge cancellation problem after the Appell transform.

8.4 The top Legendre-loop amplitude

The coefficient of x^{2N} in $P_{2N}(x)$ is

$$[x^{2N}]P_{2N}(x) = 2^{-2N} \binom{4N}{2N}. \quad (97)$$

The deconvolution operator preserves the leading coefficient. Therefore

$$A_{2N,N} = (2N)! 2^{\binom{2N+1}{2}} \lambda_{2N} 2^{-2N} \binom{4N}{2N}. \quad (98)$$

Theorem 8.3 (Exact quadratic logarithmic growth of the top block). *The top amplitudes satisfy*

$$\limsup_{N \rightarrow \infty} \frac{\log |A_{2N,N}|}{N^2} = 2 \log 2. \quad (99)$$

Proof. Since $|P_{2N}(x)| \leq 1$ on $[-1, 1]$ and $\int u = 1$, $|\lambda_{2N}| \leq (4N+1)/2$. Thus the Legendre coefficients are at most polynomially large, while nonanalyticity implies $\limsup_{N \rightarrow \infty} |\lambda_{2N}|^{1/(2N)} = 1$; otherwise an exponentially decaying coefficient bound would give analytic continuation. Hence $\log |\lambda_{2N}| = o(N)$ along the limsup subsequence. Stirling's formula gives $\log(2N)! = \mathcal{O}(N \log N)$ and $\log \binom{4N}{2N} = \mathcal{O}(N)$. The sole quadratic term in (98) is

$$\binom{2N+1}{2} \log 2 = (2N^2 + N) \log 2.$$

Divide by N^2 . □

Conjecture 8.4 (Top-block dominance). *For all sufficiently large N —possibly for every $N \geq 1$ —*

$$\max_{0 \leq n \leq 2N} |A_{n,N}| = |A_{2N,N}|. \quad (100)$$

Consequently the maximum amplitude would also have logarithmic rate $2 \log 2$. Exact computations verify (100) for $1 \leq N \leq 16$.

8.5 Analyticity obstruction to a stable fixed-coefficient loop

The finite loop has a triangular array $A_{n,N}$. Could there be fixed coefficients c_n such that $u = \sum c_n G_n(x+2)$ in a stable manner? The following theorem rules out every normally convergent version.

Theorem 8.5 (No exponentially weighted normal closure). *Suppose (c_n) satisfies*

$$\sum_{n=0}^{\infty} |c_n| R^n < \infty \quad (101)$$

for some $R > 4$. Then

$$f(z) = \sum_{n=0}^{\infty} c_n G_n(z+2) \quad (102)$$

converges normally in a complex neighborhood of $[-1, 1]$ and is analytic there. Hence it cannot agree with u on a nonempty interval.

Proof. Since $|Z| \leq 1$,

$$|G_n(z+2)| = |\mathbb{E}(z+2-Z)^n| \leq (|z+2|+1)^n.$$

Choose $\varepsilon > 0$ with $4 + \varepsilon < R$. On $|z| \leq 1 + \varepsilon$ the right-hand side is at most $(4 + \varepsilon)^n$. The Weierstrass test gives normal convergence and analyticity. Equality with u would make u real analytic on $[-1, 1]$, contradicting its nowhere-analytic character. □

Remark 8.6. The theorem does not exclude a violently conditional fixed-coefficient series in a weaker topology. It says that any such representation must live outside every natural exponentially weighted coefficient space and cannot justify interchange of the triangular sums by ordinary normal convergence.

Conjecture 8.7 (Fixed-index drift). *For every fixed $n \geq 0$, the sequence $A_{n,N}$ is unbounded as $N \rightarrow \infty$, and for many n it changes sign infinitely often. The exact data through $N = 16$ show strong drift, but a proof appears to require a quantitative nonanalyticity theorem for the Legendre coefficients rather than only the qualitative limsup used above.*

8.6 Endpoint asymptotics and a Lambert–W saddle

Equation (28) turns a two-ray endpoint ratio into a moment ratio. Since

$$G_n(y) = \int_{-1}^1 (y-t)^n u(t) dt, \quad y \in [1, 3], \quad (103)$$

the dominant contribution for large n comes from $t = -1$. Writing $t = -1 + s$ and $a = y + 1$ gives, to first Laplace order,

$$G_n(y) \approx a^n \int_0^\delta \exp\left(-\frac{n}{a}s\right) F(s) ds. \quad (104)$$

The small- s logarithm of F has a quadratic logarithmic leading term and a periodic Lambert correction in the repository's endpoint analysis [9]. Balancing the Laplace term with the derivative of the leading endpoint exponent gives the saddle scale

$$s_{n,y} \asymp \frac{a}{n \log 2} W_0\left(\frac{n \log 2}{a}\right). \quad (105)$$

This complements the lower-branch W_{-1} coordinate used when inverting F at a small value. For the next statement write

$$\mathcal{L}_F(c) = \int_0^1 e^{-cs} F(s) ds, \quad c > 0. \quad (106)$$

Only the endpoint $s = 0$ contributes to its all-orders large- c expansion.

Conjecture 8.8 (All-orders two-ray transfer asymptotics). *For fixed $y \in (1, 3]$, the ratio $G_n(y)/G_n(1)$ admits an all-orders expansion obtained by evaluating the repository's periodic Lambert endpoint profile at the pair of saddle phases $W_0(n \log 2/(y+1))$ and $W_0(n \log 2/2)$. The leading term has the form*

$$\log \frac{G_n(y)}{G_n(1)} = n \log \frac{y+1}{2} + \log \frac{\mathcal{L}_F(n/(y+1))}{\mathcal{L}_F(n/2)} + o(1), \quad (107)$$

with explicit Bell-polynomial corrections. Through (28), this would yield a uniform asymptotic theory for pairs of adjacent dyadic Fabius rays.

9 Arithmetic and q-structural consequences

9.1 Rationality

If all nodes are rational, then every $L_j^{(q)}(-2)$ is rational and every γ_{2s} is rational, so (35) gives

$$a_{nj} \in \mathbb{Q}. \quad (108)$$

This extends the rational coefficient phenomenon of the common-scale endpoint basis to the explicit multiscale cardinal ladder.

9.2 Prime support on the equispaced grid

Take

$$\xi_j = -1 + \frac{2j}{d}, \quad 0 \leq j \leq d. \quad (109)$$

Define the finite prime set

$$\mathcal{P}_d = \{p \text{ prime} : p \leq d+1\} \cup \bigcup_{1 \leq m \leq \lfloor d/2 \rfloor} \{p : p \mid 2^{2m} - 1\}. \quad (110)$$

Theorem 9.1 (Prime-support bound). *For the equispaced grid (109), every denominator of every reduced amplitude a_{nj} has prime factors only in $\mathcal{P}_d$.*

Proof. The equispaced cardinal can be written as an integer polynomial divided by $2^d j!(d-j)!$, so its coefficients and all derivatives introduce only primes at most d . By (7), the denominator of κ_{2m} can involve primes dividing m , primes dividing $2^{2m} - 1$, and primes in the denominator of B_{2m} . The von Staudt–Clausen theorem says the latter satisfy $p-1 \mid 2m$, hence $p \leq 2m+1 \leq d+1$. The recurrence (11) introduces no new denominator primes, so the same support bound holds for γ_{2m} . Apply (35). $\square$

The accompanying exact computations show, for example, that the least common denominator at $d = 8$ has prime support $\{2, 3, 5, 7, 17\}$; the prime 17 enters through $2^8 - 1$.

Problem 9.2 (Exact denominator valuations). *Determine v_p of the least common denominator of the entire matrix (a_{nj}) on equispaced, Chebyshev-rational, and dyadic grids. The theorem above gives the possible primes, but substantial cancellation occurs and the exact exponents appear to combine factorial valuations, von Staudt–Clausen data, and the factors $2^{2m} - 1$ in a nontrivial max-plus fashion.*

9.3 q-products, aliases, and the two inverse finite products

The repeated-integration coefficients use

$$\prod_{\ell=0}^{m-1} (1 - q^{2^\ell})^{-1}, \quad (111)$$

whereas common-scale alias elimination uses the annihilator $A_d(z) = \prod_{m=1}^d [2^m]_z$. Equation (55) shows that they are reciprocal sides of the same lacunary Thue–Morse calculus: integration inverts dyadic differences, while the common-scale recurrence applies them after removing the polynomial pole at $z = 1$. This gives a concrete operator-level bridge among the Lagrange coefficients, q-integers, Thue–Morse signs, and the sinc-product zero multiplicities.

10 A parity-adapted two-sided ladder

The one-sided ladder is anchored at -2 and has support extending far to the right. Even functions such as u and the Legendre partial sums admit a symmetric alternative.

Define

$$\Phi_n^L(x) = G_n(x+2) = D_n U_n(x), \quad \Phi_n^R(x) = G_n(2-x) = \Phi_n^L(-x). \quad (112)$$

For $r \geq 0$, put

$$E_r(x) = \frac{1}{2} (\Phi_{2r}^L(x) + \Phi_{2r}^R(x)). \quad (113)$$

Proposition 10.1 (Even Appell ladder). *The polynomial E_r is even, monic, and has degree $2r$. Hence $E_0, \dots, E_N$ form a basis of the even polynomials of degree at most $2N$. In particular there are unique coefficients $\eta_{r,N}$ such that*

$$S_N(x) = \sum_{r=0}^N \eta_{r,N} E_r(x). \quad (114)$$

Each E_r is a four-atom symmetric up block, with the $r = 0$ repetitions combined through the partition identity.

Proof. Reflection makes E_r even. Both summands have leading terms x^{2r} , so the average is monic. A monic degree-ordered family is triangular with respect to $1, x^2, \dots, x^{2N}$ and therefore forms a basis. $\square$

Conjecture 10.2 (Symmetric conditioning advantage). *After a natural normalization by the leading coefficient, the symmetric basis (113) has a smaller coefficient norm for the even Legendre partial sums than the one-sided basis $G_n(x+2)$. More strongly, it is asymptotically optimal among convex combinations of causal and anticausal dyadic ladders. This is plausible because it removes odd coordinates generated solely by the off-center anchor, but a proof requires sharp singular-value estimates for the two triangular bases.*

11 Exact and numerical experiments

11.1 Reproducibility protocol

The archive includes `experiments.py`. It uses exact SymPy rational arithmetic for moments, reciprocal moments, cardinal identities, Legendre coefficients, and every finite reconstruction. Floating point is used only for condition numbers and plotting. The main command is

```
python experiments.py --matrix-degree 18 \
    --legendre-order 16 --denominator-degree 8
```

The script performs the following checks.

1. It generates μ_n and γ_n from the cumulant recurrence and checks the known initial values.
2. It verifies symbolically that every quadratic cardinal residual in (32) is the zero polynomial and checks $A_{\Xi}B = I$ exactly.
3. It constructs the exact Legendre partial sums through degree 32, applies $M(D)^{-1}$, verifies every closed loop as a polynomial identity, checks the exact top-block formula, and records top-block dominance through $N = 16$.
4. It compares equispaced and Chebyshev–Lobatto Appell–Vandermonde matrices through degree 18.
5. It records exact denominator factorizations for equispaced grids through degree 8.

11.2 Conditioning of the Appell–Vandermonde factor

Figure 1 shows $\log_{10} \kappa_2(A_{\Xi})$. Chebyshev–Lobatto nodes usually improve the condition number, as expected from interpolation geometry, although both matrices become difficult at moderate degrees.

The full sample-to-atom map includes the dyadic diagonal D_n . Its norm is shown in Figure 2. The difference between node families is modest compared with the universal normalization growth predicted by Theorem 8.1.

11.3 Legendre-loop coefficient growth

Figure 3 plots $\log_{10} \max_n |A_{n,N}|$. In all exact computations through $N = 16$, the maximum occurs at the top block $n = 2N$, consistent with Theorem 8.4. The value at $N = 16$ is approximately 199.72.

At the same time, the partial sums converge on the target interval. Figure 4 shows the two exact pointwise diagnostics $|S_N(0) - u(0)| = |S_N(0) - 1|$ and $|S_N(1) - u(1)| = |S_N(1)|$. The nonmonotone but decreasing envelope is typical of spectral projection, while the atom amplitudes grow by nearly two hundred decimal orders.

Warning 11.1 (Do not evaluate the closed loop naively). The exact identities are mathematically valid, but ordinary double-precision summation of coefficients near 10^{200} is not a sensible numerical method for recovering a number of order one. The loop is an algebraic and structural representation, not a stable evaluation algorithm in its raw normalization. Exact arithmetic, compensated high precision, symmetric renormalization, or a separately conditioned common-scale nullspace compression is required.

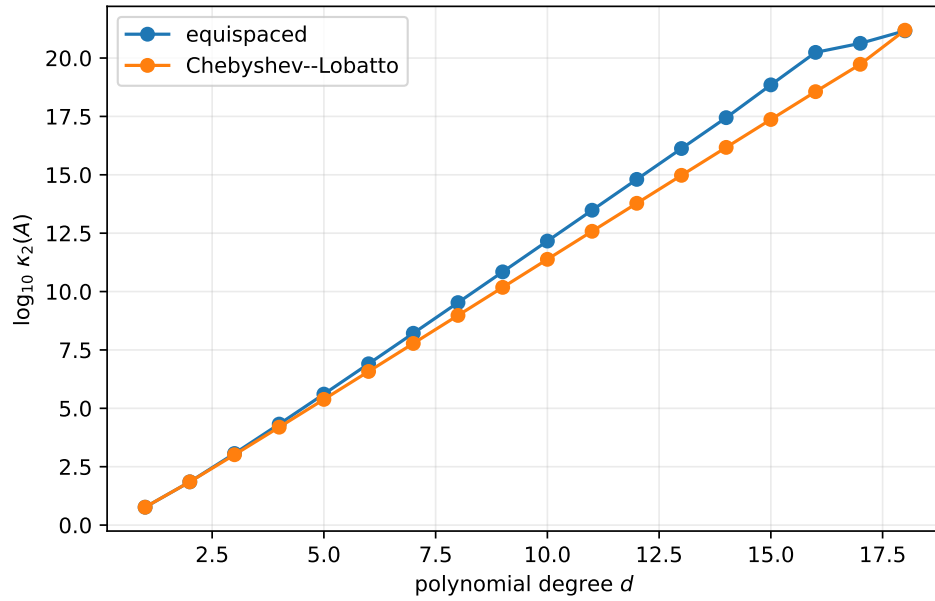


Figure 1: Conditioning of the Appell–Vandermonde matrix $A_{in} = G_n(\xi_i + 2)$. Exact formulas define the matrix; the plotted condition numbers are floating-point diagnostics.

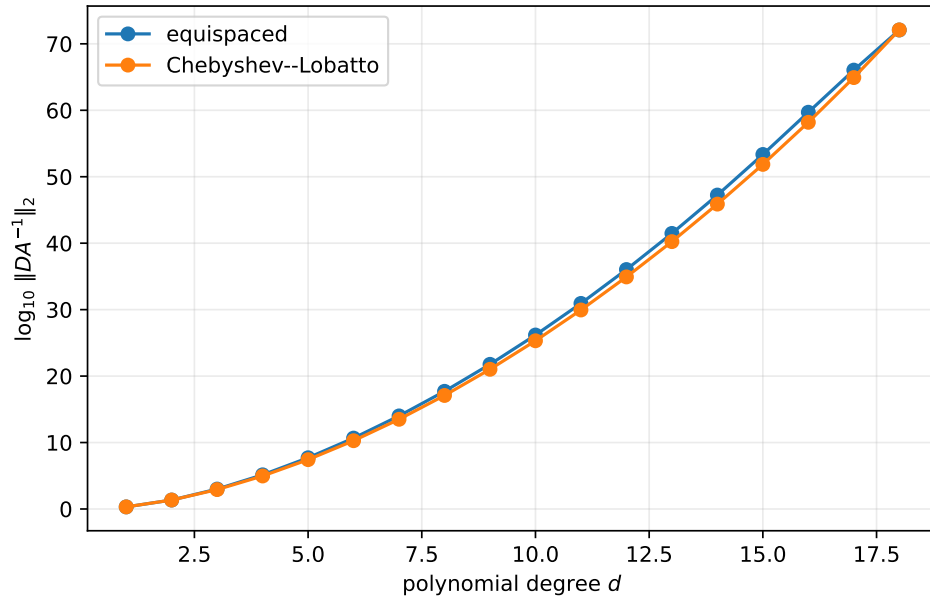


Figure 2: Norm of the raw sample-to-atom transform $\text{diag}(D_n)A_{\Xi}^{-1}$. The roughly quadratic curve on the log scale is intrinsic to the dyadic ladder.

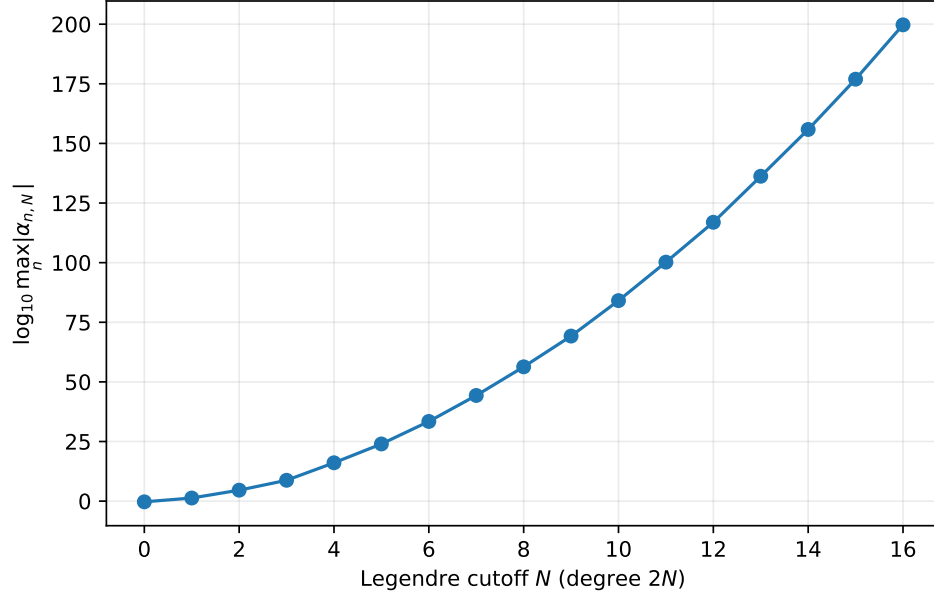


Figure 3: Growth of the largest exact raw atom amplitude in the degree- $2N$ Fourier–Legendre closure.

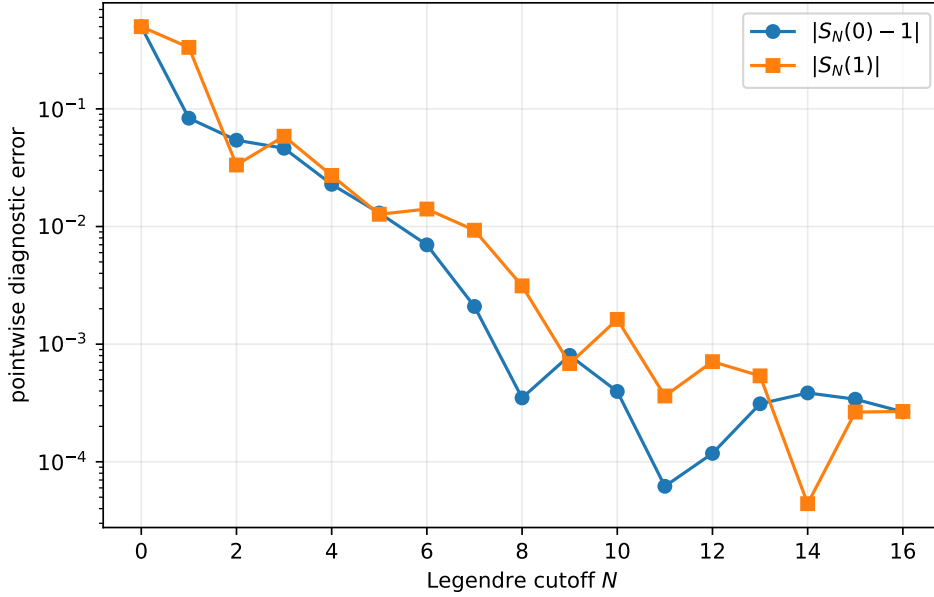


Figure 4: Pointwise convergence diagnostics for the Legendre partial sums, contrasted with the coefficient growth in Figure 3.

Table 3: Selected exact-loop diagnostics. The amplitude column is $\log_{10} \max_n |A_{n,N}|$.

N	degree	amplitude log	$S_N(0)$	$S_N(1)$
0	0	−0.3010	0.5000000	0.5000000
2	4	4.5958	1.0541667	0.0333333
4	8	16.0771	1.0228446	−0.0271723
8	16	56.3175	0.9996510	−0.0031297
12	24	116.9261	0.9998821	−0.0007100
16	32	199.7205	0.9997341	0.0002680

12 Further frontier directions

12.1 Renormalized atom bases

The factor $D_n = n!2^{\binom{n+1}{2}}$ is forced by repeated integration, but it is not the only possible normalization. Define normalized blocks

$$\hat{U}_n(x) = \frac{G_n(x+2)}{\|G_n(\cdot+2)\|_{[-1,1]}}, \quad (115)$$

or use the endpoint value $G_n(1)$ from (27). This moves the endpoint asymptotics into the basis and may reveal a more stable coefficient geometry.

Problem 12.1 (Optimal diagonal renormalization). *Find diagonal weights w_n minimizing the worst-case condition number of $\text{diag}(w_n)A_{\Xi}^{-1}$ over node sets or over Chebyshev/Fekete node families. Determine whether a Bernoulli–Bell or Lambert-based weight can reduce the quadratic-exponential growth to ordinary exponential growth.*

12.2 A topology for a fixed-coefficient closure

Theorem 8.5 excludes normal convergence but leaves open weaker possibilities.

Problem 12.2 (Nonnormal Appell expansion). *Does there exist a fixed sequence (c_n) such that $\sum c_n G_n(x+2)$ converges pointwise, distributionally, or in a Denjoy–Carleman topology to u on $[-1, 1]$? If so, characterize the weakest natural coefficient space in which the synthesis map is continuous. If not, identify the precise quasi-analytic obstruction.*

A promising route is to regard the degree-dependent vectors $(c_{0,N}, \dots, c_{2N,N})$ as a projective system in an ultradistribution space rather than insist on coordinatewise convergence.

12.3 Sharper Legendre coefficient asymptotics

The exact dyadic formula (75) and the endpoint Lambert expansion should jointly determine the sign and magnitude of λ_{2m} at a finer scale than the qualitative nonanalyticity argument. This would attack Theorems 8.4 and 8.7 and could produce a periodic modulation in the top amplitudes after removing $2N^2 \log 2$ and the Stirling terms.

Conjecture 12.3 (Periodic modulation of the top amplitude). *After subtracting the explicit elementary contribution from (98), the residual*

$$\log |A_{2N,N}| - \log \left((2N)! 2^{\binom{2N+1}{2} - 2N} \binom{4N}{2N} \right) = \log |\lambda_{2N}| \quad (116)$$

contains a bounded or slowly varying log-periodic component inherited from the dyadic endpoint phase. Establishing this would connect the spectral coefficients directly to the periodic Lambert correction.

12.4 Optimal cardinal grids for atomic extension

Maximizing the ordinary Vandermonde determinant favors Fekete/Chebyshev nodes, but the actual extension norm depends on the anchor -2 , the reciprocal moments, and the flat tails of u .

Problem 12.4 (Atomic Fekete nodes). *For each d , minimize a norm such as*

$$\|T_{\Xi}\|_2, \quad \max_j \sum_n |a_{nj}|, \quad \text{or} \quad \sup_{x \notin [-1,1]} \sum_j |\tilde{L}_j(x)| \quad (117)$$

over all node sets $\Xi_d \subset [-1, 1]$. Determine the limiting node distribution and compare it with the arcsine/Fekete law. The external anchor suggests a small but systematic asymmetry may appear for the one-sided ladder, whereas the parity-adapted ladder should restore symmetry.

12.5 Formal verification

The core identities are particularly suitable for Lean or another proof assistant. Their ingredients are finite polynomial algebra after importing four analytic lemmas: the functional-differential equation, compact support, the repeated-integration identity, and the moment generating function. A formalization can proceed in the following order:

1. formalize the recurrence for $p_m(N)$ and prove (20);
2. derive (25);
3. implement truncated formal power series for M and $1/M$;
4. prove (32) by polynomial operator inversion;
5. prove the Appell–Vandermonde determinant by a unit-triangular change of basis; and
6. derive the finite Legendre loop by substitution.

The exact rational examples and Python-generated identities provide a useful test oracle but are not substitutes for proof.

13 Conclusion

The finite synthesis problem has an explicit answer. Every Lagrange cardinal polynomial on $[-1, 1]$ is a finite sum of dyadically shifted and scaled Rvachev atoms, with coefficients obtained by applying the reciprocal moment operator at the exterior anchor -2 . Equivalently, it has a common-scale pure-translate representation and an optimal $d + 2$ -atom endpoint compression.

Substitution closes the loop in two complementary senses. A convergent Lagrange interpolation scheme for u becomes a uniformly convergent sequence of finite self-representations. The repository’s Fourier–Legendre expansion yields an even more explicit closure in which every coefficient is a finite combination of dyadic Fabius values, Legendre derivatives, Bernoulli cumulants, and Bell-polynomial reciprocal moments.

The loop is exact but not numerically benign. Its sample-to-atom determinant contains the cubic exponent $\binom{d+2}{3}$, and the top amplitude of the degree- $2N$ Legendre closure has quadratic logarithmic rate $2 \log 2$. This instability is not an accident: the transform extrapolates to -2 , where Legendre polynomials grow exponentially, and any normally convergent fixed-coefficient Appell closure would contradict the nowhere-analytic nature of u .

The resulting picture is therefore a useful duality. Polynomial approximation is stable and rapidly convergent on the physical interval, while its exact Rvachev atomization exposes an extremely delicate cancellation geometry near the flat dyadic edge. That geometry is where the Thue–Morse product, q-integer aliases, Bernoulli–Bell deconvolution, and Lambert– W endpoint saddle all meet.

A Formula sheet

For quick reference, the central formulas are collected here.

Moment data.

$$M(z) = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j},$$

$$\kappa_{2m} = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)},$$

$$\frac{1}{M(z)} = \sum_{r \geq 0} \gamma_r \frac{z^r}{r!}.$$

Two-atom Appell block.

$$U_n(x) = u\left(-1 + \frac{x+3}{2^{n+1}}\right) + u\left(-1 + \frac{x+1}{2^{n+1}}\right),$$

$$G_n(x+2) = n! 2^{\binom{n+1}{2}} U_n(x).$$

Lagrange cardinal coefficients.

$$b_{nj} = \frac{1}{n!} \sum_{r=0}^{d-n} \frac{\gamma_r}{r!} L_j^{(n+r)}(-2),$$

$$a_{nj} = n! 2^{\binom{n+1}{2}} b_{nj},$$

$$L_j(x) = \sum_{n=0}^d a_{nj} U_n(x).$$

Matrix form.

$$A_{in} = G_n(\xi_i + 2),$$

$$(b_{nj}) = A_{\Xi}^{-1},$$

$$\det T_{\Xi} = \frac{2^{\binom{d+2}{3}} \prod_{n=0}^d n!}{\prod_{i < j} (\xi_j - \xi_i)}.$$

Legendre loop.

$$S_N = \sum_{m=0}^N \lambda_{2m} P_{2m},$$

$$A_{n,N} = 2^{\binom{n+1}{2}} \sum_{m=\lceil n/2 \rceil}^N \lambda_{2m} \sum_{r=0}^{2m-n} \frac{\gamma_r}{r!} P_{2m}^{(n+r)}(-2),$$

$$S_N(x) = \sum_{n=0}^{2N} A_{n,N} U_n(x),$$

$$u(x) = \lim_{N \rightarrow \infty} \sum_{n=0}^{2N} A_{n,N} U_n(x).$$

B Data files in the archive

The ZIP archive contains:

- `lagrange_up_loop.tex`: this report;
- `lagrange_up_loop.pdf`: the rendered report;
- `experiments.py`: exact symbolic and numerical experiments;
- `README.md`, `requirements.txt`, and `SHA256SUMS`: reproduction instructions and integrity data;

- `data/node_conditioning.csv`;
- `data/legendre_loop_summary.csv`;
- `data/legendre_loop_coefficients.csv`;
- `data/amplitude_denominators.csv`; and
- the four PDF figures used in Section 11.

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